SongCluster

A model for automatic Spotify playlist generation

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Introducing our Model

SongCluster

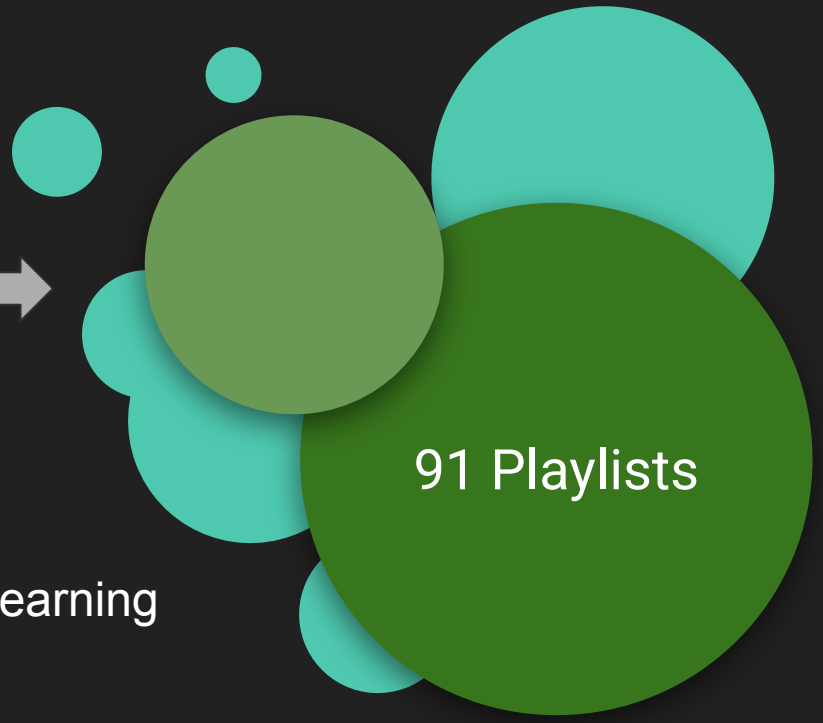
- Prototype model
- Creates Spotify playlists based on mood
- Quick, efficient categorisation of large numbers of songs

Here, we present our model and some of the playlists it has created

SongCluster

We tested our model on a Spotify dataset

- 5171 songs
- 2176 artists



How does it work?

- Clustering using unsupervised machine learning

Our Data

The playlists are based on song features provided directly by Spotify.

Included Features:

tempo 🥁 ⏱️

energy ⚡ 🤸

danceability 🥳 🪩

Excluded Features:

artist name 🕺 💃

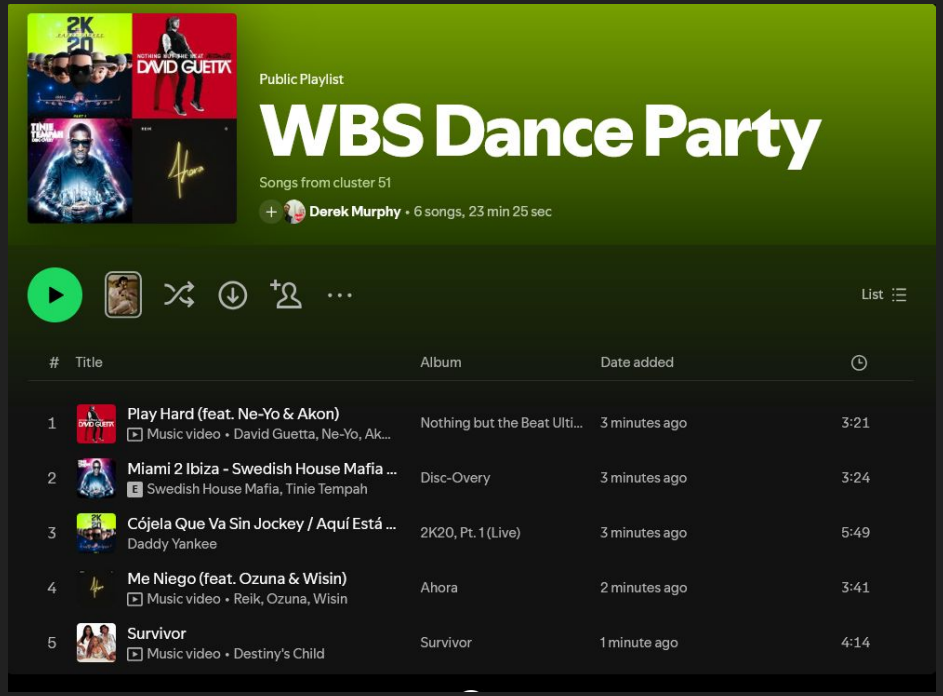
key 🎼 🎵

html 🌐 💻

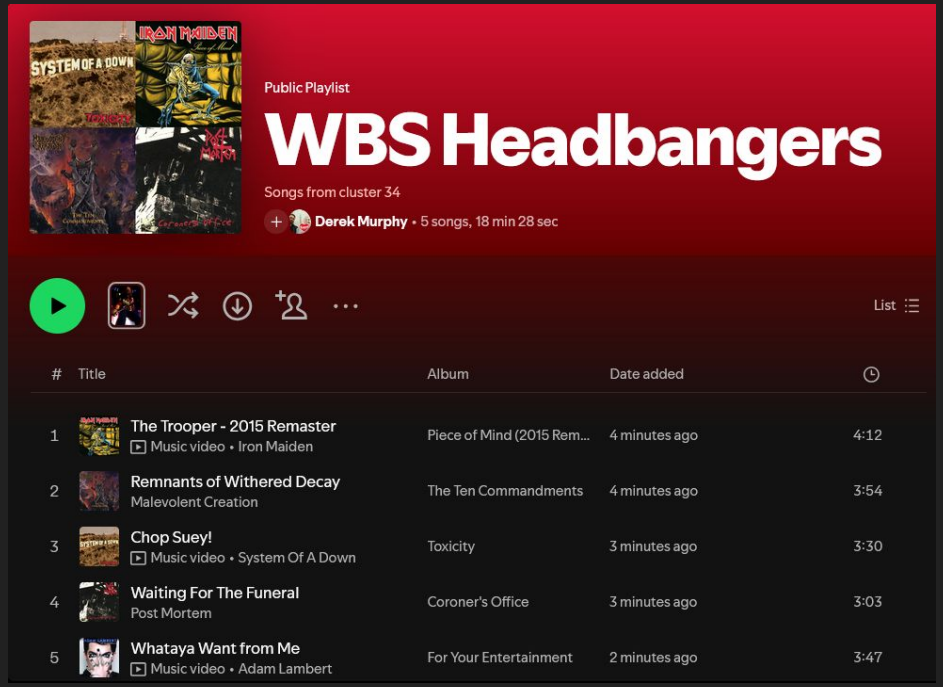
Exploring the playlists

Playlist 51

- Danceable, high energy, loud, upbeat songs
- Hip hop, reggaeton, electronica, eurodance
- ‘Dance party!’ 💃🕺💃🕺



Exploring the playlists



Playlist 34

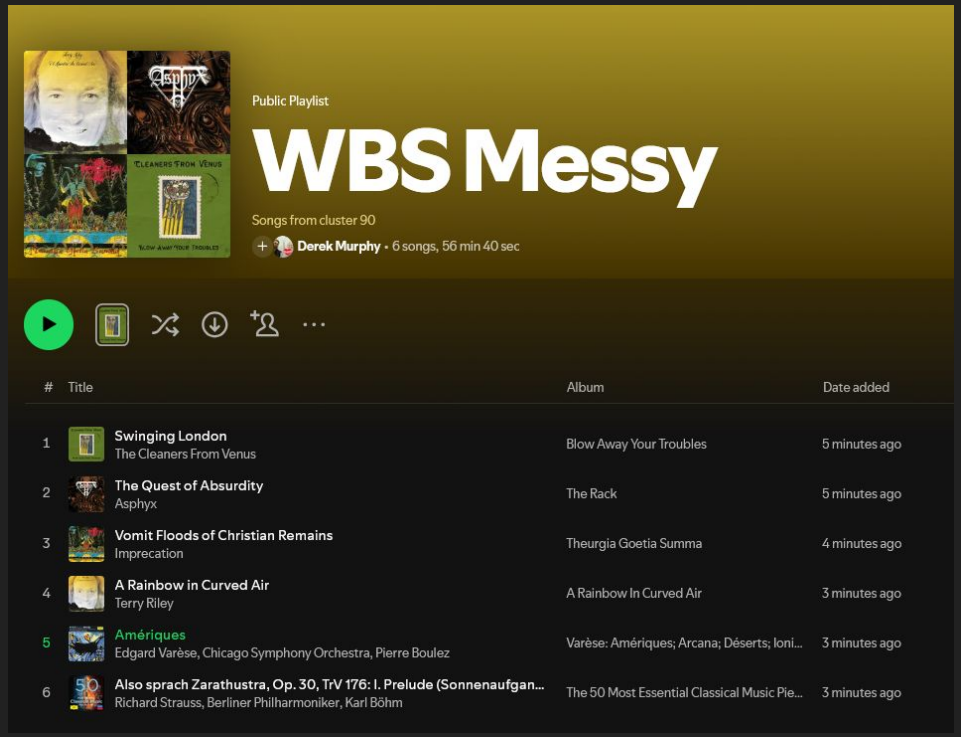
- High energy, very loud, dark songs with low danceability
- Death metal, Nu-Metal, Alternative, Pop-Rock
- ‘Headbangers!’ 💀🎸🤘

Exploring the playlists

Playlist 7

- slow, instrumental, low energy, quiet
- Ambient, classical, instrumental
- ‘Calm & Meditative!’ ☯️🧘‍♀️🙌🏻😌

Exploring the playlists



It doesn’t always work well

Playlist 90

- Mostly dark and moody but high variability in energy, tempo
- Ambient, Death Metal, Classical, Indie
- ‘Messy’ 🤯🤔

Additional Signals for Better Playlist Generation

🎼 **Genre**

- Example: Alternative Metal → “Chop Suey”

🎧 **User Listening Behaviour**

- Example: Reggaeton fans → “Play Hard“

📝 **Lyrics / Mood**

- Example: Calm / melancholic → “Beethoven 250“

🎤 **Artist’s Style**

- Example: Electronic artists → “David Guetta“

🎯 **Audio Features + Human Perception → Better Playlist Generation**

Why SongCluster?

✅ Pros

- Fast and scalable
- Compatible with data from Spotify
- Produces interpretable song groupings
- Flexible number of playlists

⚠️ Cons

- Difficult to ensure a perfect playlist fit
- Cannot control playlist size
- Requires data pre-processing

Next steps?

- Expand the dataset to the entire Spotify database
- More features → better classification
- Label playlists by mood
- Human-curated playlists with ML support